

Question Paper

Exam Date & Time: 18-Nov-2025 (02:00 PM - 05:00 PM)



MANIPAL ACADEMY OF HIGHER EDUCATION

MANIPAL INSTITUTE OF TECHNOLOGY - BENGALURU
SEVENTH SEMESTER B.TECH. DEGREE EXAMINATIONS, NOVEMBER - 2025

Text Analytics [CSE_4006]

Marks: 50

Duration: 180 mins.

PART-A

Answer all the questions.

- 1) Demonstrate the document standardization process with an example. List the advantages of the same. (5)

A)

Course Outcome: "CO1" / Cognitive Level: "Understand"

- B) An organization receives thousands of customer feedback emails every month. These emails cover different topics such as product quality, delivery issues, customer service, and pricing. The company wants to analyze them efficiently to identify common themes and improve decision-making. (5)

a) Analyze how document clustering can be applied in this scenario to group similar feedback together.

b) Describe how organizing the clustered documents can help the organization improve its data analysis and response process.

Course Outcome: "CO1" / Cognitive Level: "Analysis"

- 2) A data analytics company is building a Natural Language Processing (NLP) system to analyze financial news headlines. Consider the example statement: (5)

A) *"The investors are buying and selling technology stocks aggressively."*

Experiment how tokenization, stemming, and lemmatization would be applied to preprocess this text. Show the transformation of the statement at each stage and discuss how these processes help improve the accuracy and performance of the NLP model used for financial text analysis.

Course Outcome: "CO2" / Cognitive Level: "Application"

- B) List the algorithm steps involved in vector generation for text data prediction using suitable example. (5)

A company is developing an AI-based news recommendation system that analyzes the meaning of sentences to suggest related financial articles. Consider the example statement:

"Stock prices are rising rapidly in the global market".

Apply vector generation to represent this sentence numerically for machine learning.

Course Outcome: "CO2" / Cognitive Level: "Application"

- 3) Apply tokenization, POS tagging, and named entity recognition to generate text features for classification. Consider the following text as an example "Apple CEO Tim Cook announced new products in California." (5)

A)

(Note: Consider only the text within the quotes)

Course Outcome: "CO3" / Cognitive Level: "Application"

- B) Identify challenges faced when applying traditional classification models to text data. How does the spreadsheet analogy explain text classification? (5)

Course Outcome: "CO3" / Cognitive Level: "Application"

- 4) Compare and contrast text Taxonomy and Ontologies. (5)

- A) A healthcare organization is building a text analytics system to analyze patient feedback and categorize it into themes such as "treatment quality," "staff behavior," and "facility issues." However, some terms like "doctor attitude" and "nurse politeness" are being treated as separate categories instead of being grouped under "staff behavior." Analyze how using an ontology could help improve the categorization of such feedback and give one advantage of applying ontology-based analysis in this context.

Course Outcome: "CO4" / Cognitive Level: "Analysis"

- B) Identify and explain the three different types of Pointwise Mutual Information types used for measuring semantic similarity. (5)

Course Outcome: "CO4" / Cognitive Level: "Application"

- 5) Compare entity detection and extraction in NLP using suitable example and list the algorithm steps involved in Sequential Tagging-Based Entity Extraction. (5)

- A) Course Outcome: "CO5" / Cognitive Level: "Analysis"

- B) Build and Explain the general architecture of Question Answering system and its uses cases. (5)

Course Outcome: "CO5" / Cognitive Level: "Application"

-----End-----

Event Name	Subject
------------	---------

058-END SEMSTER-ODD-2025 CSE_4006:Text Analytics

#	Course Outcome	Description	Marks
1	1	CO 1	10
2	2	CO 2	10
3	3	CO 3	10
4	4	CO 4	10
5	5	CO 5	10
Total			50

* **NA** - CO Not Selected

Event Name	Subject
------------	---------

058-END SEMSTER-ODD-2025 CSE_4006:Text Analytics

#	Level	No of Questions	Marks (%)
1	Remember	0	0(0%)
2	Understand	1	5(10%)
3	Application	6	30(60%)
4	Analysis	3	15(30%)
5	Create	0	0(0%)
6	Evaluate	0	0(0%)
Total		10	50 (100%)